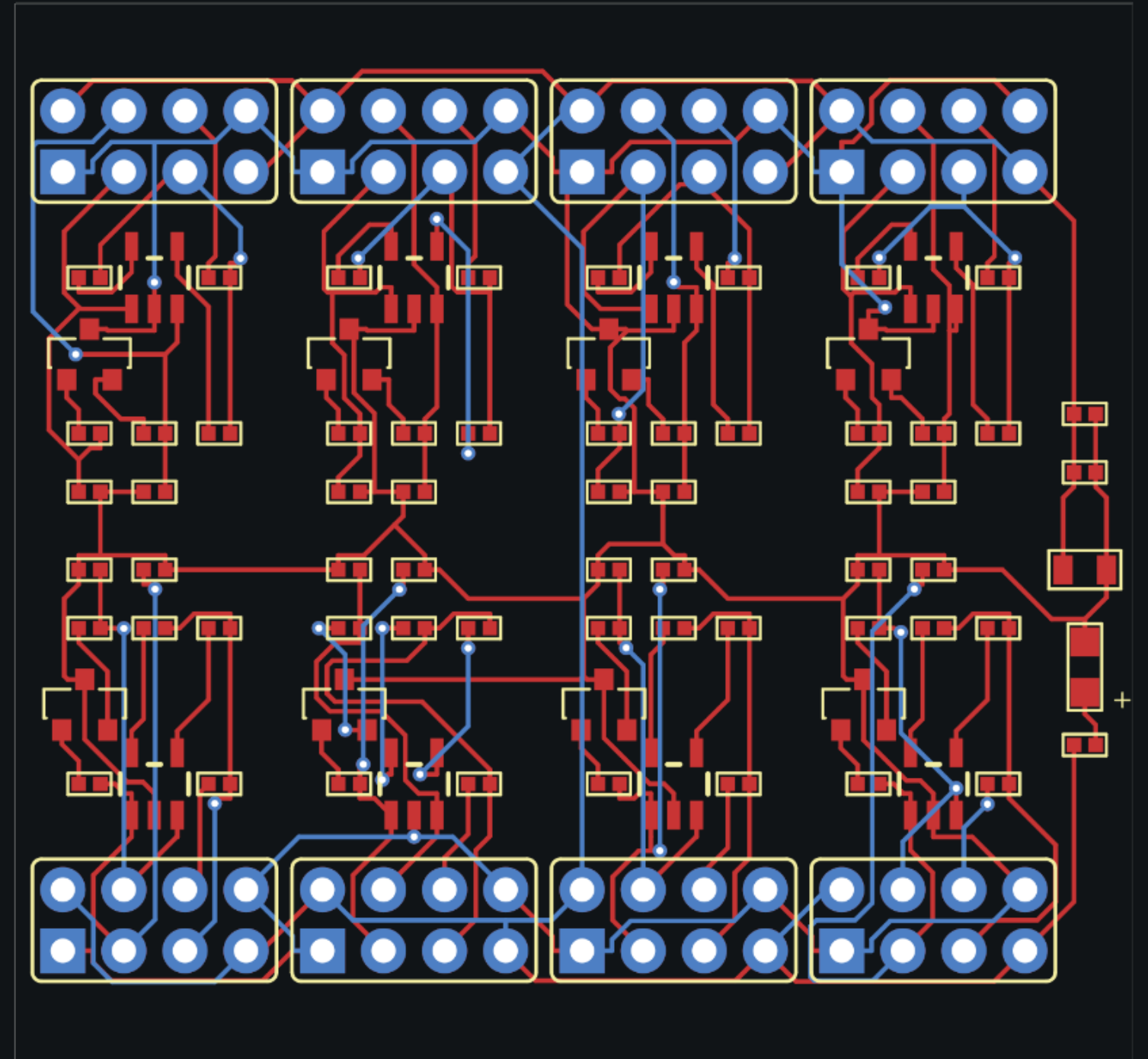


PCB Loop

AI-guided PCB
placement optimization

Philippe Ferreira De Sousa

CoreWeave Hackathon



Recorded Medium candidate

Placement and routing loop

Stage 1: reach validity

Preserve the circuit and manufacturing constraints

Stage 2: lower the score



One persistent task can run a candidate or a batch, then revise the experiment

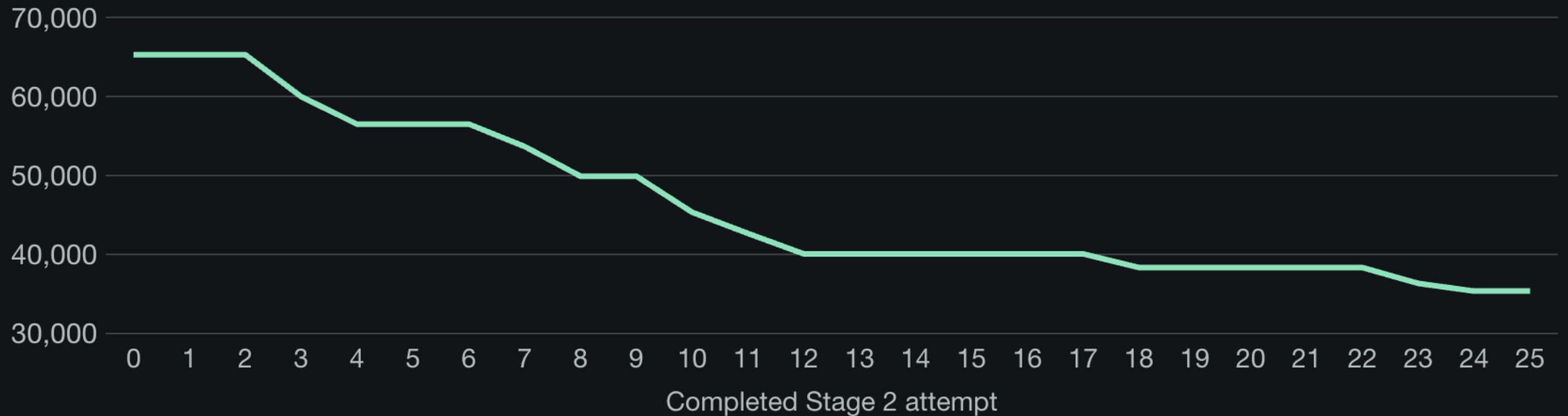
Medium score fell by 45.8%

65,288

Starting valid board

35,365

Best recorded valid board

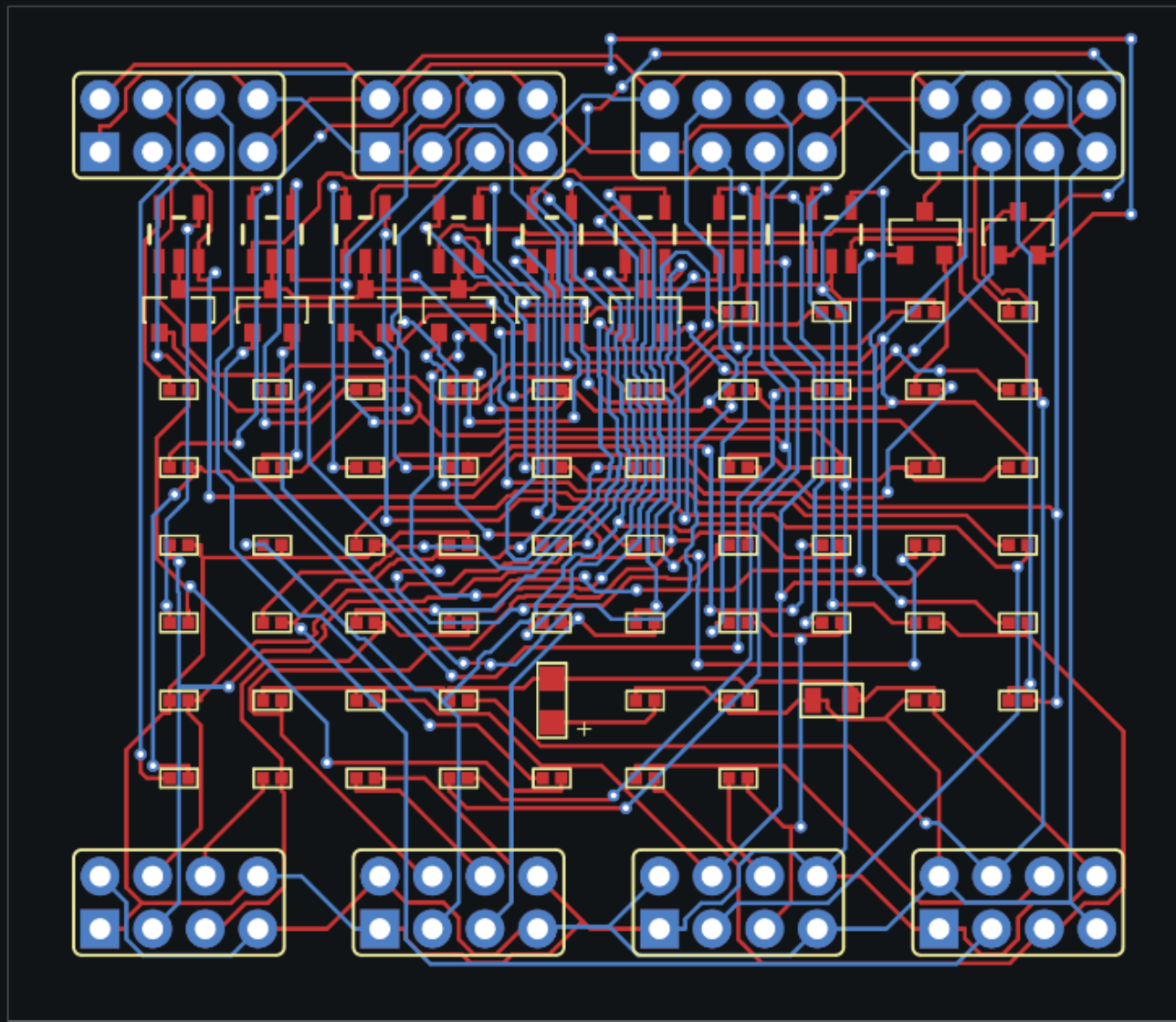


Score = assembled volume (mm³) + 50 × vias + 5,000 × copper layers

85 components, reduced demo circuit. Original/full board remains incomplete.

Functional grouping reduced the via cost

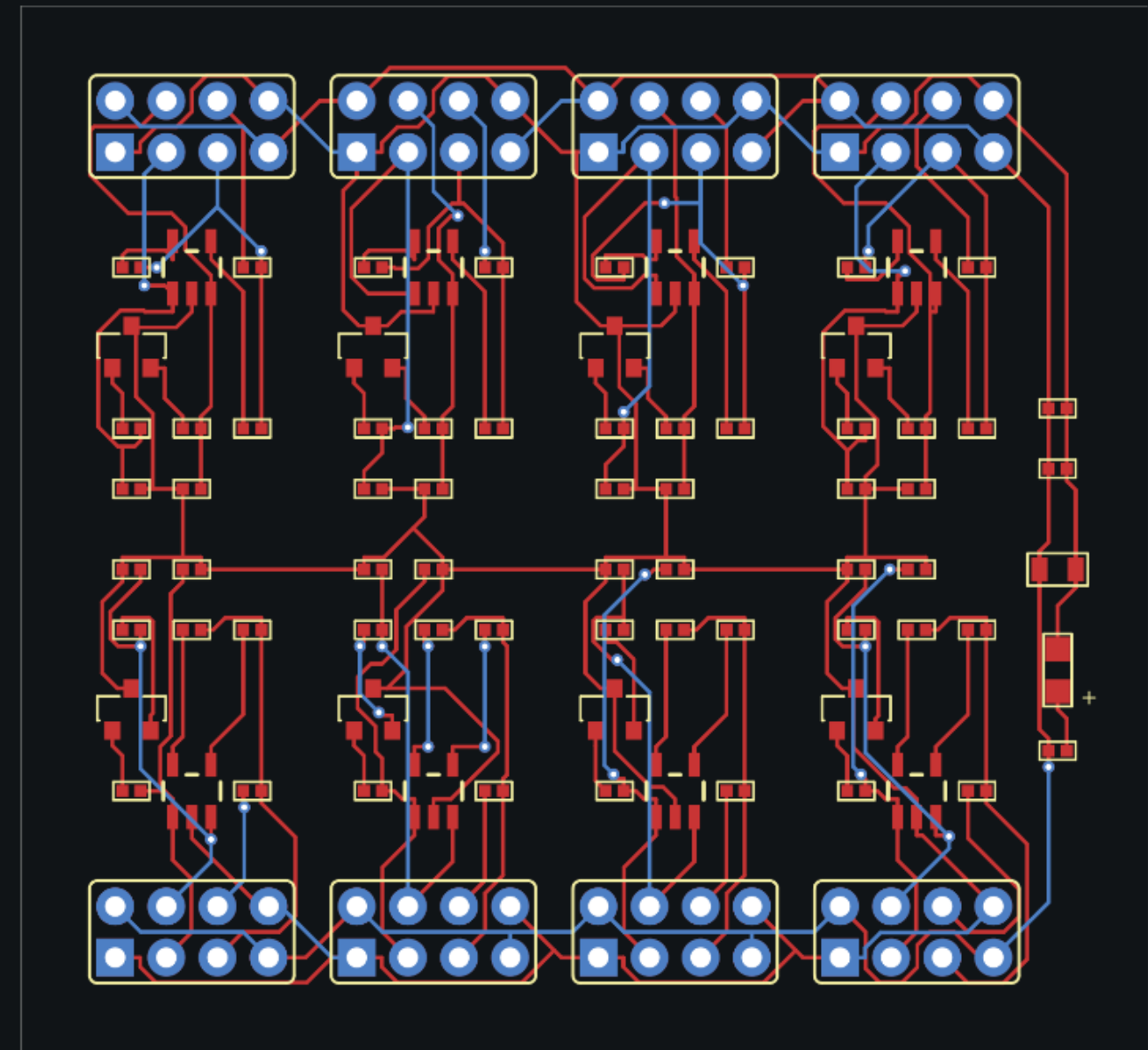
Before grouping



148 vias

Score 49,910.52

Connected blocks near their headers



29 vias

Score 45,345.68

W&B connects each result to its evidence

Evidence	What I can inspect
Charts	Valid score, vias, volume and rejected attempts
Weave record	Action parameters, commands and validation output
Artifacts	Candidate PCB, native reports and source hashes

Open the optimization run

Open the grouping trace

Completed experiment records. Routing time comes from command logs.

W&B upload verified through 40,074.08. The latest local result is 35,365.44.

Results and next step

Board family	Components	Recorded status
Small	15	Valid reduced circuit
Medium	85	Valid, 45.8% lower score
Large	156	Incomplete
Original / Full	245	Incomplete

Next: qualify the original 245-component board

Public code, prompts and candidate history

github.com/PhilippeFerreiraDeSousa/copper-scar